Resolution Meets Importance Sampling: An Empirical Study of Sample Count and Gradient Estimation in Variational Image Models

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Abstract*—Importance-weighted variational objectives and doubly reparameterized gradient (DReG) estimators are almost always benchmarked on binarized images, yet the observation model is itself a modelling choice that controls the variance of the importance weights those methods depend on. We ask four questions: whether raising observation complexity from binary to 16-level to full grayscale systematically changes density-estimation complexity; whether DReG-IWAE consistently improves on standard IWAE in bound quality and optimization stability; whether increasing the number of importance samples K gives consistent gains or diminishing returns; and whether these effects hold across image content. Using one fixed 784–200–200–z_{20}–200–200 architecture, we quantize the same MNIST images to $R \in \{2, 16, 256\}$ levels and pair each with an exact discrete likelihood (Bernoulli, Categorical, discretized logistic) so all conditions are comparable in bits per dimension. A protocol frozen by a pass/fail pilot gate spans $R \times K \in \{1, 5, 10, 20, 50\} \times$ {IWAE, DReG} over three seeds (81 runs), repeated identically on EMNIST-Letters (81 runs), with every checkpoint scored at a common $K_{\text{eval}} = 1000$. Complexity rises by 0.47 and 0.98 bpd per resolution step; DReG wins 23 of 24 matched cells but its stability advantage is resolution-dependent; K helps roughly six times more in absolute bits at $R=256$ than at $R=2$ and saturates later; and the three headline effects replicate across both datasets.

***Index Terms*—variational inference, importance-weighted autoencoder, doubly reparameterized gradients, latent-variable models, density estimation, Monte Carlo, bits per dimension**

I. Introduction

Deep latent-variable models specify a joint density $p_\theta(x, z) = p(z)p_\theta(x \mid z)$ whose marginal likelihood $p_\theta(x) = \int p_\theta(x, z)\, dz$ is intractable. The variational autoencoder (VAE) [1] makes learning possible by maximizing the evidence lower bound (ELBO) under an amortized approximate posterior $q_\phi(z \mid x)$, and the importance-weighted autoencoder (IWAE) [2] tightens that bound by averaging K importance weights inside the logarithm. Tighter bounds are not free: the standard reparameterized gradient of the K-sample bound carries a score-function term whose signal-to-noise ratio for the inference network decays as K grows [3], which the doubly reparameterized gradient (DReG) estimator [4] removes at the cost of an extra backward pass.

Almost every empirical statement about these mechanisms has been made on *binarized* MNIST. Binarization is a modelling decision, not a property of the data, and it fixes the observation model to the lowest-entropy likelihood one could attach to an image. Because the importance weight $w = p_\theta(x, z)/q_\phi(z \mid x)$ inherits its scale and spread almost entirely from $\log p_\theta(x \mid z)$, that decision controls the very quantity IWAE and DReG exist to manage. A diagnostic on our own initialized models makes the point: the mean log weight at initialization is -558 nats under a Bernoulli likelihood, -2215 under a 16-way Categorical, and -4749 under a discretized logistic over 256 levels. Conclusions calibrated on binary images need not transfer to the resolutions people actually deploy.

We therefore ask four questions on a common bits-per-dimension scale. **RQ1:** does raising observation complexity from binary to 16-level to full grayscale cause a systematic change in density-estimation complexity? **RQ2:** does DReG-IWAE consistently improve on standard IWAE, in gradient stability as well as in density-estimation quality? **RQ3:** does increasing the number of importance samples K give consistent gains, or do we eventually observe diminishing returns? **RQ4:** using both MNIST digits and EMNIST characters, do these effects remain consistent across different types of image content? The corresponding hypotheses are that bits per dimension rises systematically with R (H1); that DReG is the better estimator on both quality and stability grounds, because it removes the score-function term whose variance grows as the weights concentrate (H2); that K gives monotone but diminishing returns (H3); and that all three effects carry over from digits to letters (H4).

Our contribution is not a new algorithm but a controlled, frozen, fully-crossed measurement of a question the IWAE/DReG literature has left unmeasured — executed on a scale that makes the three observation conditions legitimately comparable, and repeated in full on a second dataset so that the *pattern* of effects, not just their endpoints, can be checked for consistency. We show that (i) density complexity rises systematically and by a large margin with observation complexity; (ii) DReG wins 23 of 24 matched cells on bound quality but its stability advantage is real only at coarse resolutions and reverses at $R = 256$, so H2 is only partially supported; (iii) the benefit of K is monotone, sharply diminishing, and roughly six times larger in absolute bits at $R = 256$ than at $R = 2$, with a saturation point that moves outward as resolution grows; and (iv) all three effects reproduce on EMNIST characters, while the fine-grained interaction contrasts do *not*, which we report rather than suppress.

II. Background and Related Work

The foundational paper for this study is Burda *et al.*'s importance-weighted autoencoder [2], which sits inside a short and unusually well-connected line of work.

The VAE [1], [6] introduced amortized variational inference with the reparameterization trick, turning posterior inference into a differentiable, low-variance Monte Carlo estimate. Its known weakness is that the ELBO penalizes the approximate posterior for mass placed where the true posterior has none, so a factorized Gaussian q_ϕ becomes over-regularized and latent units collapse to the prior. Burda *et al.* diagnosed this as a property of the *bound* rather than the architecture and replaced the ELBO with a K-sample importance-weighted bound $\mathcal{L}_K$, monotone non-decreasing in K and convergent to $\log p_\theta(x)$. Because $\mathcal{L}_K$ only requires the proposal to cover the regions carrying weight, it relaxes the pressure on q_ϕ and, as they showed empirically, increases the number of active latent units.

A tighter bound does not automatically give a better gradient. Rainforth *et al.* [3] showed that the signal-to-noise ratio of the standard reparameterized inference-network gradient of $\mathcal{L}_K$ scales as $O(1/\sqrt{K})$: as the bound tightens the normalized weights concentrate, the useful signal shrinks faster than its noise, and the encoder is trained by an increasingly uninformative gradient. Roeder *et al.* [5] had already observed that dropping the explicit score-function path through q_ϕ ("sticking the landing") removes a variance term that vanishes at the optimum. Tucker *et al.* [4] generalized both observations: applying the reparameterization a second time, to the weights themselves, yields the DReG estimator, in which the inference gradient is weighted by $\tilde{w}_k^2$ instead of $\tilde{w}_k$ and the explicit ϕ-path through $\log q_\phi$ is detached. DReG is unbiased for the same objective and shares the same forward samples, which makes it an unusually clean experimental factor: a DReG-vs-IWAE comparison holds model, samples and weights fixed and varies only the backward pass.

A second, largely separate literature concerns *what* the decoder should predict. Theis *et al.* [7] showed that log-likelihoods computed under different observation models are not mutually comparable, and that continuous densities fitted to discrete pixels can be inflated arbitrarily; the remedy is to model the discrete pixel mass directly and report bits per dimension. Salimans *et al.* [8] popularized the discretized logistic likelihood for exactly this reason: it respects the ordinal structure of intensity and integrates the underlying continuous density over the correct bin. That construction is what makes our resolution factor measurable at all.

These literatures have not been crossed. IWAE and DReG are evaluated on binarized MNIST [2], [4]; discretized likelihoods are studied in autoregressive and flow models where K-sample bounds play no role, and work on richer posteriors [9] attacks posterior mismatch architecturally rather than through the estimator. Our study occupies the gap: architecture fixed, observation model treated as a controlled factor, and the estimator-side conclusions tested against the change.

III. Probabilistic Model

A. Variables and generative story

Let $u \in \{0, \dots, 255\}^D$ with $D = 784$ be the raw 8-bit image. For a resolution $R \in \{2, 16, 256\}$ we deterministically quantize

$$q_i^{(R)} = \mathrm{round}\Big(\frac{u_i}{255}(R-1)\Big), \qquad x_i^{(R)} = \frac{q_i^{(R)}}{R-1}, \tag{1}$$

so $q^{(R)} \in \{0, \dots, R-1\}^D$ is the *observed random variable* (the modelled quantity) and $x^{(R)} \in [0,1]^D$ is the network input. For $R = 256$, $q = u$ exactly. All three conditions are generated from the *same* byte tensor, so any R-effect is a property of the representation and not of a different data sample.

The latent variable is $z \in \mathbb{R}^{20}$ with prior $p(z) = \mathcal{N}(0, I)$. The generative model factorizes as

$$p_\theta(q, z) = p(z) \prod_{i=1}^{D} p_\theta(q_i \mid z), \tag{2}$$

i.e. pixels are conditionally independent given z — the classical latent-variable conditional-independence assumption, with z acting as the single d-separating node between all 784 observed variables. θ denotes decoder parameters and ϕ encoder parameters; both are model parameters, not random variables (this is a maximum-likelihood, not a fully Bayesian, treatment).

B. Observation models

The likelihood is the only R-dependent part of the model (Table I). For $R = 2$ each pixel is Bernoulli with logit $\ell_i(z)$; for $R = 16$ each pixel is Categorical over 16 levels with logits $\ell_{i,\cdot}(z)$. For $R = 256$ we use a single discretized logistic: with $\tilde{x}_i = 2q_i/255 - 1$, location $\mu_i(z) = \tanh(\cdot)$, scale $s_i(z) = \mathrm{softplus}(\cdot)$ and half-bin width $\delta = 1/255$,

$$p_\theta(q_i \mid z) = \sigma\left(\frac{\tilde{x}_i + \delta - \mu_i}{s_i}\right) - \sigma\left(\frac{\tilde{x}_i - \delta - \mu_i}{s_i}\right), \tag{3}$$

with the edge bins absorbing the tails ($q_i = 0$ uses the lower CDF, $q_i = 255$ the upper). Equation (3) is a genuine

probability *mass*: it integrates a continuous logistic density over the bin that the observed intensity actually occupies, which is what keeps it comparable to the Bernoulli and Categorical cases [7]. We verified numerically that the 256 bin masses sum to one to within 2.0×10^{-6}.

Because the three likelihoods live on different alphabets, all comparisons use bits per dimension,

$$\mathrm{bpd}(q) = \frac{-\log_2 p_\theta(q)}{D} = \frac{-\log p_\theta(q)}{D \ln 2}, \tag{4}$$

the average number of bits needed to code one pixel. This is the single choice that makes RQ1 answerable.

C. Objectives

The inference model is the diagonal Gaussian $q_\phi(z \mid x) = \mathcal{N}(z; \mu_\phi(x), \mathrm{diag}\, \sigma^2_\phi(x))$, sampled by reparameterization $z = \mu_\phi + \sigma_\phi \odot \epsilon$, $\epsilon \sim \mathcal{N}(0, I)$, so that gradients pass through the sample. Exact inference of $p_\theta(z \mid q)$ is intractable because the normalizer $p_\theta(q)$ is a 20-dimensional integral against a neural likelihood; all inference here is approximate.

The $K = 1$ baseline maximizes the ELBO with the analytic Gaussian KL,

$$\mathcal{L}_{\mathrm{VAE}} = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(q \mid z)] - D_{\mathrm{KL}}(q_\phi(z \mid x) \,\|\, p(z)) \,. \tag{5}$$

The first term rewards reconstruction; the second penalizes posterior mass that the prior does not support, and is the term responsible for latent collapse.

For $K > 1$ we draw $z_1, \ldots, z_K \overset{iid}{\sim} q_\phi(z \mid x)$, form weights $w_k = p_\theta(q, z_k)/q_\phi(z_k \mid x)$, and maximize

$$\mathcal{L}_K = \mathbb{E}_{z_{1:K}}\left[\log \frac{1}{K}\sum_{k=1}^{K} w_k\right], \qquad \log p_\theta(q) \geq \mathcal{L}_{K+1} \geq \mathcal{L}_K. \tag{6}$$

$\mathcal{L}_K$ is the log of an unbiased Monte Carlo estimate of the marginal likelihood; Jensen's inequality makes it a lower bound, and the monotonicity in (6) is Burda *et al.*'s Theorem 1 [2]. Probabilistically, (6) asks q_ϕ only to *cover* the posterior well enough that K draws contain one good one, instead of asking it to *be* the posterior.

D. Gradient estimators

Writing $\tilde{w}_k = w_k / \sum_j w_j$ for the normalized weights, the standard reparameterized gradient of (6) is

$$\nabla_{\{\theta,\phi\}} \mathcal{L}_K \approx \sum_{k=1}^{K} \tilde{w}_k \, \nabla_{\{\theta,\phi\}} \log w_k, \tag{7}$$

where the ϕ-part contains both a pathwise term (through z_k) and an explicit score-function term $-\nabla_\phi \log q_\phi(z_k \mid x)$. As K grows the weights concentrate on one sample, the pathwise signal shrinks, and the residual score term dominates — the $O(1/\sqrt{K})$ SNR decay of [3].

DReG [4] keeps (7) for θ but replaces the inference gradient by

$$\nabla_\phi \mathcal{L}_K^{\mathrm{DReG}} = \sum_{k=1}^{K} \tilde{w}_k^2 \, \frac{\partial \log w_k}{\partial z_k} \frac{\partial z_k}{\partial \phi}, \tag{8}$$

i.e. the explicit dependence of $\log q_\phi$ on ϕ is detached (only the pathwise dependence through z_k survives) and the weights enter squared. Two consequences matter here. The squared weights make the estimator's variance a function of how *dispersed* the weights are, which is precisely the quantity our resolution factor manipulates; and because the forward pass, the samples and the weights are shared with (7), an IWAE-vs-DReG comparison isolates the backward pass alone.

E. Evaluation, sampling and uncertainty

At test time every model, whatever it was trained with, is scored by the same estimator: (6) with $K_{\text{eval}} = 1000$, computed by exact `logsumexp` over memory-safe chunks of 50, then converted with (4). This decouples training-time tightness from evaluation-time tightness. The reported quantity is a finite-sample *lower bound* on $\log p_\theta(q)$, not the exact NLL; we name it accordingly throughout.

Sampling is direct in both places it is needed: ancestral sampling $z \sim p(z)$, $q \sim p_\theta(\cdot \mid z)$ for generation, and importance sampling $z_{1:K} \sim q_\phi$ for training and evaluation. There is no MCMC, so cost is $O(K)$ decoder evaluations per example.

Four sources of uncertainty are distinguished: *posterior* uncertainty, represented explicitly by q_ϕ and summarized by the KL term; *observation* uncertainty, carried by the likelihood head (the logistic scale s_i is a learned, input-dependent uncertainty); *Monte Carlo* uncertainty in the bound, controlled by K during training and pinned at $K_{\text{eval}} = 1000$ at test time; and *optimization* uncertainty across initializations, measured by three seeds and summarized with bootstrap intervals. We also track *active units* (latent coordinates whose posterior-mean variance over the validation set exceeds 10^{-2}), Burda *et al.*'s diagnostic for whether the latent space is used at all.

IV. Methodology

A. Data

The primary dataset is MNIST [10]: the official 60,000 training images are split once, with a fixed split seed, into 50,000 train and 10,000 validation; the official 10,000-image test set is sealed and never used for tuning, checkpoint selection, or configuration selection. The generalization dataset is EMNIST [11], split `letters`: 104,000 train, 20,800 validation, 20,800 test, 26 balanced classes. Preprocessing is exactly (1) — no binarization sampling, no augmentation, no normalization beyond the division by $R - 1$.

B. Architecture

The trunk is fixed across all conditions: $784 \to 200 \to 200 \to \{\mu, \log\sigma\} \in \mathbb{R}^{20}$, and $20 \to 200 \to 200 \to$ head, with $\tanh$ activations and Xavier-uniform initialization [12]. Only the head differs (Table I). A single stochastic layer is used, deliberately simpler than Burda *et al.*'s two-layer variant, since depth is not a factor of this study.

Table I exposes a confound we did not remove: the 16-level Categorical head is $6.8\times$ larger than the Bernoulli head. Equalizing capacity would have meant deviating from the

TABLE I
OBSERVATION MODEL AND COST PER RESOLUTION CONDITION. ONLY THE DECODER HEAD CHANGES; THE TRUNK IS IDENTICAL. HEADS ARE SIGMOID, SOFTMAX AND (tanh LOCATION, SOFTPLUS SCALE) RESPECTIVELY. MACS ARE ANALYTIC DENSE-LAYER COUNTS FOR ONE EXAMPLE AT $K = 1$.

R	Likelihood	Par./pixel	Params	MACs
2	Bernoulli	1	407,224	405,600
16	Categorical(16)	16	2,770,984	2,757,600
256	Disc. logistic	2	564,808	562,400

TABLE II
FROZEN PROTOCOL. NO VALUE WAS RE-TUNED AFTER THE PILOT GATE.

Setting	Value
Optimizer	Adam [13], $\beta = (0.9, 0.999)$, $\epsilon = 10^{-4}$
Learning rate	10^{-3}, constant
Batch size / epochs	128 / 100
Latent dim. d_z	20 (hard-locked, never swept)
Weight init.	Xavier uniform
Precision / TF32	float32 / disabled, deterministic kernels
Checkpoint selection	best validation bpd at training K
Evaluation	$K_{\text{eval}} = 1000$, chunk 50, exact logsumexp
Active-unit threshold	10^{-2}
Bootstrap	10,000 resamples, seed 424242
Seeds	$\{1, 2, 3\}$

frozen protocol, so we quantify the imbalance and discuss it in Section VII. What is *not* confounded is worth stating: the encoder is identical across all three conditions (204,800 linear MACs per example in every case), so the inference network on which IWAE and DReG actually operate is the same network throughout, and only the decoder differs (200,800, 2,552,800 and 357,600 MACs per latent sample). We also verified the quantization: each condition realizes exactly R distinct levels spanning 0 to $R - 1$, and the mean normalized intensity is stable at 0.1717 and 0.1719 for $R = 16$ and $R = 256$ but falls to 0.1664 at $R = 2$ — binarization is the one step that visibly distorts the marginal statistics of the data.

C. Training protocol and the pilot gate

All numeric choices (Table II) were frozen *before* the main grid ran, gated by a pilot with predeclared pass/fail thresholds rather than by a tuning sweep. Ten pilot runs ($R \in \{2, 256\}$, $K \in \{1, 10, 50\}$, both estimators, seed 1) were checked for (i) late-window convergence: mean validation improvement over the last 10 epochs below 0.005 bpd; (ii) late-window stability: no worsening beyond 0.010 bpd; (iii) gradient spikes: per-batch gradient-norm maximum below $100\times$ the median; and (iv) finiteness of every logged quantity. All ten passed, so the protocol was frozen unchanged; had any failed, the rule was to edit the configuration, document it and rerun the pilot, never to adjust a hyperparameter in response to a main-grid or test outcome.

Validation uses common random numbers across epochs, so the epoch-to-epoch differences driving checkpoint selection reflect the model rather than Monte Carlo noise. Runs are stored in immutable directories keyed by a hash of the implementation, protocol, dataset bytes and software environment, so a resumed run cannot silently mix two protocols.

D. Baselines

Two baselines answer two different questions. The $K = 1$ VAE (5) is the reference for RQ3 ("does importance weighting help, and by how much, at this resolution?"); the standard IWAE gradient (7) is the reference for RQ2 ("is the DReG backward pass worth it?"), and is the strictest comparison available: same objective, same forward samples, same weights, same seed-paired initialization.

E. Computational setup

PyTorch 2.11 (CUDA 13.0) [14] on a single NVIDIA RTX 4080 SUPER (16 GB), 28 logical CPU cores, 31 GB RAM. The grid manifest declares 172 runs (81 MNIST, 81 EMNIST, 10 pilot); all 172 completed and both missing-run lists are empty; all 63 result artifacts carry SHA-256 digests alongside software, environment and dataset fingerprints. Pure training time totalled ≈ 6.6 GPU-hours for MNIST and ≈ 13.7 for EMNIST, excluding validation and the $K_{\text{eval}} = 1000$ evaluations. The pilot's extrapolated estimate for the MNIST grid (6.0 GPU-hours) was within 10% of the measured value. Every cell of both grids completed: the expected-grid manifest reports zero missing runs, and all 162 runs share one software, dataset and benchmark-environment fingerprint, so no result was produced under a different code path or data version. The released bundle contains 63 artifacts with SHA-256 digests, including the frozen train/validation index arrays.

V. EXPERIMENTAL DESIGN

The design is fully crossed: $R \in \{2, 16, 256\} \times \big[K = 1\ (\text{VAE}) + K \in \{5, 10, 20, 50\} \times \{\text{IWAE, DReG}\}\big] \times 3$ seeds $= 27$ configurations $\times\ 3 = 81$ runs per dataset. At $K = 1$ the two estimators are identical by construction, so that cell is run once as the VAE baseline.

The primary metric is test bpd at $K_{\text{eval}} = 1000$. Secondary measurements are the reconstruction term, the analytic KL term (both in bpd, so the decomposition is additive and comparable across R), the active-unit count, the selected epoch, wall-clock time to the selected checkpoint, total training time, and peak GPU memory.

Stability (RQ2) is measured in two complementary ways. Direct per-batch gradient diagnostics — the maximum-to-median gradient-norm ratio and a finiteness check — were collected during the pilot, where they form part of the pass/fail gate; they were deliberately switched off for the 162 main runs because the required device synchronization would have contaminated the wall-clock measurements. For the main grid we therefore use the across-seed standard deviation of the final test bpd and the width of the paired bootstrap intervals, which measure how reproducibly an estimator lands on the same solution from different random initializations.

With three seeds per cell a factorial ANOVA's assumptions cannot be checked, so we predeclared a descriptive plan: mean $\pm$ SD per configuration and paired nonparametric

bootstrap intervals (10,000 resamples) [15] for every contrast, including as the strictest RQ4 check the three-way contrast $\Delta_{RKE} = \left[\Delta^{\text{DReG–IWAE}}_{R_{\text{hi}}} - \Delta^{\text{DReG–IWAE}}_{R_{\text{lo}}}\right]_{K_{\text{hi}}} - \left[\cdot\right]_{K_{\text{lo}}}$. All intervals are exploratory summaries, not confirmatory tests. Saturation follows a predeclared reporting convention: the first K after which all later adjacent gains fall below 10% of the initial $K=1\to5$ gain.

Beyond the main grid we report three additional analyses: an estimator ablation (IWAE vs. DReG at every R, K), a computational-efficiency analysis (performance against training cost and memory), and the matched EMNIST grid, released only after the complete MNIST analysis was finished so that it could not influence it.

VI. Results

A. RQ1: observation complexity

Table III shows the expected monotone ordering: on MNIST the best model reaches 0.1324 bpd at $R = 2$, 0.6028 at $R = 16$ and 1.5787 at $R = 256$. Matched at equal K and estimator, the $2 \to 16$ gap is 0.471–0.475 bpd and the $16 \to 256$ gap is 0.975–1.010 bpd, with bootstrap intervals far from zero in every one of the 18 comparisons. H1 is supported, and the effect is an order of magnitude larger than every other effect in this study — which is why bpd, and not raw nats, is the only defensible common currency here.

The interesting part is that the gap is not a constant. As K grows from 1 to 50 the $16 \to 256$ gap *shrinks* from 1.0104 to 0.9754 bpd on MNIST and from 1.6327 to 1.6016 on EMNIST; the $2 \to 16$ gap shrinks correspondingly from 0.4750 to 0.4708 and from 0.7069 to 0.6989. Importance weighting buys back part of the price of finer resolution, and it buys back more of it where the price is higher, so the answer to RQ1 is not a single constant but a systematic shift whose size depends mildly on how the model is trained. We return to that coupling under RQ4.

B. RQ2: DReG versus the standard IWAE gradient

DReG produces the lower bpd in 11 of 12 MNIST cells and 12 of 12 EMNIST cells (Table IV, Fig. 2). The paired bootstrap interval excludes zero in 8/12 MNIST and 11/12 EMNIST cells. The typical advantage is 1–2×10^{-3} bpd at $R \in \{2, 16\}$ — small in absolute terms, but consistent, and comparable in size to the entire gain from $K = 10$ to $K = 50$ at those resolutions.

The improvement is consistent but it is *flat in* K rather than growing. Averaged over resolutions the MNIST advantage is $-1.51, -0.61, -2.29, -1.07$ ($\times 10^{-3}$ bpd) at $K = 5, 10, 20, 50$: reliable in sign, non-monotone in size. On EMNIST it is flat at $R \in \{2, 16\}$ but does grow at $R = 256$ ($-3.86 \to -5.17$ from $K = 5$ to $K = 50$). Consistency of *direction* is therefore much better established than any claim about how the margin scales.

On stability the answer is more qualified. All ten pilot runs passed the predeclared gradient guard — every logged quantity finite, no clipping needed, no run approaching the maximum-to-median gradient-norm ratio of 100 — and neither estimator showed a numerical-stability failure anywhere in the study. The main-grid proxy (Table V) then splits by resolution: DReG is the more reproducible estimator at $R = 2$ and $R = 16$ but the *less* reproducible one at $R = 256$ on both datasets, giving it the smaller seed SD in only 14 of 24 matched cells.

Convergence separates the two more cleanly, and not in DReG's favour. Paired contrasts on the *selected epoch* show no systematic difference (-4.0 to $+5.3$ epochs, intervals excluding zero in 5 of 24 cells), whereas contrasts on *wall-clock time* to that same checkpoint are positive in all 24 cells and exclude zero in 23, from $+21$ s to $+739$ s: DReG needs no fewer optimization steps, and its entire extra cost is per-step. One axis does favour it without qualification — counting per-seed active units, DReG reaches all 20 of 20 latent coordinates in every one of its 12 MNIST configurations, IWAE in 11 of 12 (18.67 at $R = 2, K = 5$), and the VAE baseline in none below $R = 256$ (15.0 at $R = 2$, 18.0 at $R = 16$).

All of this measures the *achieved optimum* under a fixed 100-epoch budget with validation-selected checkpoints, not the gradient signal-to-noise ratio itself, which is the quantity Tucker *et al.*'s theory speaks to.

C. RQ3: importance samples and diminishing returns

Every adjacent-K step improves the bound except one negligible -0.01×10^{-3} bpd step (MNIST, $R = 2$, DReG, $10 \to 20$), so the gains are consistent in sign, matching the theoretical monotonicity of (6); but they diminish sharply: the $K=1\to5$ step is 5–10$\times$ larger than any later step (Table VI, Fig. 1).

The magnitude of the benefit scales with resolution. Going from the VAE to DReG at $K = 50$ improves MNIST bpd by 0.0072 at $R = 2$, 0.0117 at $R = 16$ and 0.0462 at $R = 256$ — a $6.4\times$ spread in absolute bits. The saturation point moves with R in the same direction: under the predeclared rule, both estimators saturate at $K = 20$ for $R = 2$ and $R = 16$ on MNIST, but neither has saturated by $K = 50$ at $R = 256$, where the final $20 \to 50$ step is still worth 5–9×10^{-3} bpd. For a practitioner the operational reading is that the "right" K is not a property of the model family but of the observation model.

D. RQ4: consistency across image content

The EMNIST-Letters grid inherits every hyperparameter, so any difference between the two datasets is attributable to image content rather than to tuning. Absolute bpd is higher everywhere (Table VII): 0.1680, 0.8668 and 2.4664 for the best configuration at $R = 2, 16, 256$, against 0.1324, 0.6028 and 1.5787 on MNIST. Letters are a harder density-estimation target, as expected.

What matters is the pattern. Because the grids are matched cell for cell, every contrast has a partner on the other dataset, so consistency can be counted rather than described (Table VIII). Across the three families answering RQ1–RQ3, 52 of 54 matched contrasts agree in sign. The resolution effect is the strongest case: all 18 agree in sign *and* all 18 are resolved on both datasets, so the RQ1 ordering is reproduced

TABLE III

MNIST MAIN GRID: TEST BITS PER DIMENSION AT $K_{\text{EVAL}} = 1000$, MEAN OVER 3 SEEDS, SD IN PARENTHESES IN UNITS OF 10^{-3} BPD; RECONSTRUCTION AND ANALYTIC-KL TERMS ALSO IN BPD. LOWER IS BETTER; BEST PER COLUMN IN BOLD.

		$R = 2$ (Bernoulli)			$R = 16$ (Categorical)			$R = 256$ (disc. logistic)		
K	Est.	bpd	recon	KL	bpd	recon	KL	bpd	recon	KL
1	VAE	0.1396 (1.34)	0.1060	0.0464	0.6146 (1.54)	0.5684	0.0677	1.6249 (1.34)	1.5977	0.0831
5	IWAE	0.1347 (0.16)	0.1094	0.0456	0.6068 (1.16)	0.5746	0.0645	1.5993 (4.15)	1.6294	0.0721
5	DReG	0.1333 (0.13)	0.1052	0.0474	0.6051 (0.52)	0.5713	0.0651	1.5978 (4.63)	1.6237	0.0721
10	IWAE	0.1341 (0.10)	0.1127	0.0452	0.6058 (0.59)	0.5815	0.0618	1.5921 (1.32)	1.6605	0.0688
10	DReG	0.1327 (0.04)	0.1099	0.0457	0.6038 (0.58)	0.5769	0.0628	1.5937 (1.97)	1.6554	0.0691
20	IWAE	0.1337 (0.35)	0.1173	0.0435	0.6050 (0.26)	0.5868	0.0603	1.5882 (1.48)	1.7159	0.0668
20	DReG	0.1327 (0.18)	0.1149	0.0442	0.6032 (0.64)	0.5836	0.0606	1.5842 (5.00)	1.6964	0.0671
50	IWAE	0.1334 (0.11)	0.1259	0.0414	0.6045 (0.75)	0.5983	0.0573	1.5793 (4.73)	1.7594	0.0640
50	DReG	**0.1324** (0.03)	0.1207	0.0430	**0.6028** (0.54)	0.5926	0.0585	**1.5787** (2.48)	1.7830	0.0655

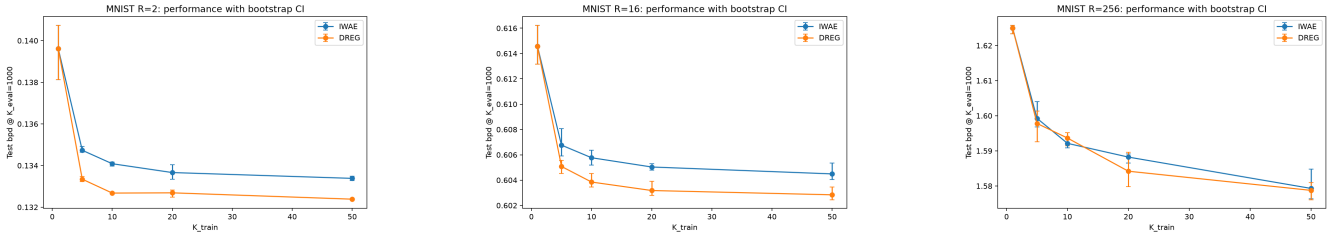


Fig. 1. MNIST test bpd at $K_{\text{eval}} = 1000$ against training K with bootstrap 95% intervals, for $R = 2$, 16 and 256 (left to right); the IWAE and DReG branches share the $K = 1$ VAE point. The vertical axes differ by an order of magnitude between panels. The curve flattens by $K \approx 20$ at $R = 2$ and $R = 16$ but is still descending at $K = 50$ for $R = 256$: this is the visual form of the saturation result in Table VI.

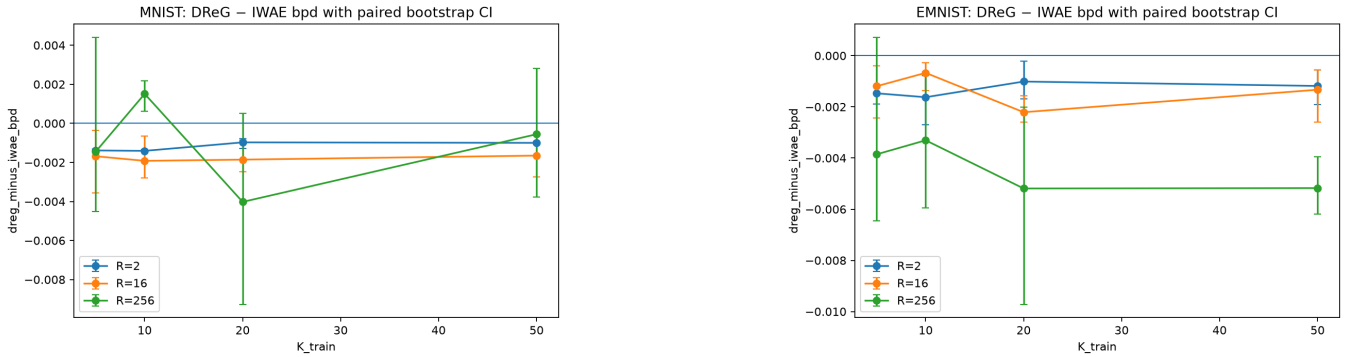


Fig. 2. DReG − IWAE test bpd against K, one series per resolution, with paired bootstrap 95% intervals, on MNIST (left) and EMNIST-Letters (right); the horizontal line is zero and points below it favour DReG. This pair is the matched RQ4 view and should be read side by side: the $R = 2$ and $R = 16$ series are almost interchangeable across datasets, whereas the $R = 256$ series is noisy and crosses zero at $K = 10$ on MNIST — the one cell where DReG loses — but separates cleanly from zero and grows with K on EMNIST.

and resolved twice over. The K-gain family agrees in 23 of 24 and the estimator effect in 11 of 12, the lone disagreement being the $R = 256, K = 10$ cell where MNIST favours IWAE.

The shapes match too: the resolution gap shrinks with K on both datasets (1.6327 $\rightarrow$ 1.6016 bpd for 16 $\rightarrow$ 256 on EMNIST, against 1.0104 $\rightarrow$ 0.9754 on MNIST), the diminishing-returns curve keeps its form with much larger gains at high resolution (33.8×10^{-3} bpd for the first step at $R = 256$ against 4.1 at $R = 2$), and the resolution-dependent stability split of Table V reproduces as well.

Two things do not carry over, and both concern fine structure rather than direction. The predeclared saturation point lands in different places, but this is largely an artifact of the rule: the threshold is 10% of the initial $K = 1 \rightarrow 5$ gain, so EMNIST's unusually small initial gain at $R = 2$ sets a threshold that later steps then exceed. More substantively, the strictest test available — the three-way $R \times K \times$estimator contrast — does not transfer. Four of six contrasts agree in sign, but *not one of the six* is resolved on both datasets, and the two largest MNIST values (-5.60 and $+3.26 \times 10^{-3}$ bpd for the $16 \rightarrow 256$ pair) shrink to -0.35 and -0.86 on EMNIST. With three seeds per cell and a quantity that is a difference of differences of differences, the noise floor is too high to support any claim about the fine-grained shape of the interaction, so we record a null result at that level. The coarse claims survive it: the magnitude of the K-benefit depends strongly on R, and the

TABLE IV

DREG − IWAE TEST BPD ($\times 10^{-3}$; NEGATIVE FAVOURS DREG) WITH PAIRED BOOTSTRAP 95% INTERVALS OVER 3 SEEDS.

		MNIST		EMNIST	
R	K	Δ	95% CI	Δ	95% CI
2	5	−1.39	[−1.45, −1.31]	−1.47	[−1.89, −1.18]
2	10	−1.41	[−1.45, −1.34]	−1.62	[−2.70, −0.78]
2	20	−0.97	[−1.28, −0.78]	−1.01	[−1.69, −0.21]
2	50	−1.00	[−1.08, −0.92]	−1.18	[−1.91, −0.55]
16	5	−1.69	[−3.56, −0.35]	−1.19	[−2.43, −0.40]
16	10	−1.92	[−2.78, −0.66]	−0.67	[−1.37, −0.28]
16	20	−1.86	[−2.47, −0.87]	−2.21	[−2.59, −1.57]
16	50	−1.65	[−2.74, −0.64]	−1.33	[−2.59, −0.55]
256	5	−1.45	[−4.51, 4.40]	−3.86	[−6.45, 0.72]
256	10	1.51	[0.61, 2.18]	−3.31	[−5.94, −0.65]
256	20	−4.02	[−9.27, 0.50]	−5.19	[−9.72, −2.02]
256	50	−0.56	[−3.77, 2.80]	−5.17	[−6.19, −3.96]

TABLE V

SECONDARY RQ2 EVIDENCE. σ: ACROSS-SEED SD OF TEST BPD ($\times 10^{-3}$, AVERAGED OVER $K \in \{5, 10, 20, 50\}$; SMALLER IS MORE REPRODUCIBLE, BETTER VALUE IN BOLD). Δt: DREG − IWAE WALL-CLOCK SECONDS TO THE SELECTED CHECKPOINT AT $K = 50$ (POSITIVE = DREG SLOWER). DREG IS STEADIER AT COARSE RESOLUTIONS AND LESS STEADY AT $R = 256$, AND IS SLOWER TO ITS BEST CHECKPOINT IN ALL 24 MATCHED CELLS.

	MNIST			EMNIST		
R	σ_{IWAE}	σ_{DReG}	Δt	σ_{IWAE}	σ_{DReG}	Δt
2	0.180	**0.094**	+20.8	**0.357**	0.358	+40.9
16	0.687	**0.571**	+333.6	0.873	**0.720**	+738.6
256	**2.920**	3.519	+134.2	**2.007**	2.300	+267.8

TABLE VI

ADJACENT-K IMPROVEMENTS IN BPD ($\times 10^{-3}$, POSITIVE = BETTER) AND THE PREDECLARED SATURATION POINT K_{SAT} ("–" = NOT REACHED BY $K = 50$).

Data	R	Est.	$1 \to 5$	$5 \to 10$	$10 \to 20$	$20 \to 50$	K_{sat}
MNIST	2	IWAE	4.88	0.65	0.42	0.28	20
MNIST	2	DReG	6.27	0.67	−0.01	0.30	20
MNIST	16	IWAE	7.82	1.00	0.73	0.54	20
MNIST	16	DReG	9.50	1.23	0.67	0.34	20
MNIST	256	IWAE	25.67	7.12	3.91	8.92	–
MNIST	256	DReG	27.12	4.16	9.45	5.46	–
EMNIST	2	IWAE	2.59	0.82	0.69	0.33	–
EMNIST	2	DReG	4.06	0.98	0.08	0.51	–
EMNIST	16	IWAE	7.51	2.03	1.29	1.53	–
EMNIST	16	DReG	8.70	1.51	2.82	0.65	50
EMNIST	256	IWAE	29.93	6.90	2.05	2.66	20
EMNIST	256	DReG	33.79	6.35	3.94	2.64	50

resolution penalty shrinks with K, on both datasets.

E. Probabilistic behaviour and efficiency

Two diagnostics explain *how* the bound improves. First, the decomposition in Table III moves in the opposite direction to naive intuition: as K grows the reconstruction term gets *worse* (at $R = 256$, from 1.598 to 1.783 bpd) while the KL term falls (from 0.083 to 0.066) — the total nevertheless improves. The K-sample objective is buying a broader, less prior-pinned posterior at the cost of single-sample reconstruction quality, exactly the behaviour (6) is supposed to produce. The active-unit counts of Section VI-B confirm the freed capacity is

TABLE VII

EMNIST-LETTERS MAIN GRID: TEST BPD AT $K_{\text{EVAL}} = 1000$, MEAN OVER 3 SEEDS (SD IN PARENTHESES, $\times 10^{-3}$ BPD).

K	Est.	$R = 2$	$R = 16$	$R = 256$
1	VAE	0.1736 (0.25)	0.8804 (0.96)	2.5132 (2.12)
5	IWAE	0.1710 (0.22)	0.8729 (0.85)	2.4832 (1.26)
5	DReG	0.1695 (0.16)	0.8717 (0.35)	2.4794 (3.41)
10	IWAE	0.1702 (0.47)	0.8709 (0.99)	2.4763 (1.87)
10	DReG	0.1685 (0.57)	0.8702 (0.52)	2.4730 (2.97)
20	IWAE	0.1695 (0.47)	0.8696 (0.78)	2.4743 (3.68)
20	DReG	0.1685 (0.28)	0.8674 (1.13)	2.4691 (2.18)
50	IWAE	0.1691 (0.27)	0.8681 (0.87)	2.4716 (1.23)
50	DReG	**0.1680** (0.42)	**0.8668** (0.88)	**2.4664** (0.63)

TABLE VIII

RQ4 CONSISTENCY, COUNTED OVER MATCHED MNIST/EMNIST CONTRAST PAIRS. "SAME SIGN" COUNTS PAIRS POINTING THE SAME WAY ON BOTH DATASETS; "BOTH RESOLVED" COUNTS PAIRS WHOSE PAIRED BOOTSTRAP INTERVALS EXCLUDE ZERO ON *both*. THE THREE FAMILIES ANSWERING RQ1–RQ3 AGREE IN 52 OF 54 PAIRS; NO THREE-WAY CONTRAST IS RESOLVED ON BOTH DATASETS.

Effect family	pairs	same sign	both resolved
Resolution effect (RQ1)	18	**18**	**18**
Adjacent-K gain (RQ3)	24	23	15
DReG − IWAE (RQ2)	12	11	9
Three-way contrast	6	4	**0**

TABLE IX

MNIST COST–BENEFIT AT THE TWO MOST EXPENSIVE RESOLUTIONS. TIMES ARE MEANS OVER 3 SEEDS FOR THE FULL 100-EPOCH BUDGET; MEMORY IS PEAK TRAINING GPU ALLOCATION.

R	K	Est.	test bpd	train (s)	best (s)	GPU (MB)
256	1	VAE	1.6249	229	252	82
256	50	IWAE	1.5793	282	311	522
256	50	DReG	1.5787	413	446	561
16	50	IWAE	0.6045	568	593	1030
16	50	DReG	0.6028	903	927	1041

used rather than wasted, reproducing Burda *et al.*'s central qualitative claim in our setting. Second, the seed SD at $R = 256$ is 2–4× that at $R = 2$, which is the empirical footprint of the higher-variance weights that the resolution factor was designed to induce.

The cost side is unambiguous (Table IX). Training time is K-dominated and DReG adds 5–59% on top: at $R = 16, K = 50$ it costs 903 s against 568 s for IWAE, to buy 1.7×10^{-3} bpd. Peak GPU memory is driven almost entirely by K (82 MB at $K = 1$ to 561 MB at $K = 50$ for $R = 256$). If a fixed wall-clock budget is the constraint rather than a fixed K, the recommendation from this table is to spend it on a larger K with the cheaper IWAE backward pass before spending it on DReG.

VII. DISCUSSION

A. Were the hypotheses supported?

H1 (density complexity rises with observation complexity) is supported decisively, though partly by construction — a 256-level target carries more information than a binary one,

so the interesting content of RQ1 was never the sign but the size, and the fact that the gap is itself a mild function of K. H2 (DReG consistently better on quality and stability) is supported on quality — 23 of 24 matched cells — but only partially on stability: DReG is the more reproducible estimator at $R \in \{2, 16\}$ and the less reproducible one at $R = 256$, and neither estimator ever triggered a numerical-stability failure, so the honest verdict is "better bound, resolution-dependent steadiness" rather than a blanket stability win. H3 (monotone, diminishing returns from K) is supported, with the important refinement that both the size of the gain and the saturation point are resolution-dependent rather than universal. H4 (consistency across image content) is supported for all three headline effects and not supported for the fine-grained interaction contrasts, which disagree across datasets.

B. Why the model behaved this way

The mechanism we believe is doing the work is weight dispersion. A 256-level discretized-logistic likelihood assigns far smaller per-pixel masses than a Bernoulli one, so $\log w$ is both larger in magnitude and more spread across samples (-558 vs. -4749 nats at initialization). A more dispersed weight distribution means, first, that the K-sample average is a higher-variance estimate of $p_\theta(x)$, so each extra sample removes more of the Jensen gap — which is why the K-gains and the saturation point both scale with R; and second, that the normalized weights concentrate faster, which is the regime DReG was built for — which is why its largest advantages and its largest run-to-run variability both appear at $R = 256$. That double effect is the most plausible reading of Table V: DReG steadies the inference gradient where the weights are mild, but at $R = 256$ the $\tilde{w}_k^2$ weighting concentrates the update on very few samples, helping the bound on average while making the trajectory more seed-sensitive. It is also why the fine-grained contrasts fail — the corner where effects are biggest is the corner three seeds resolve worst.

The reconstruction/KL inversion is the clearest probabilistic signature in the results. A reader looking only at reconstruction error would conclude that $K = 50$ damages the model at $R = 256$; in bound terms it does the opposite. The K-sample objective does not ask the posterior to make any single z reconstruct well, it asks the ensemble to cover, and the active-unit counts confirm that the freed-up posterior capacity is used rather than wasted.

C. Failure cases and limitations

We report four failures honestly. *One cell contradicts the trend*: at $R = 256, K = 10$ on MNIST, DReG is worse than IWAE by 1.51×10^{-3} bpd with an interval excluding zero. With $n = 3$ it is as likely a seed artifact as a real inversion, and we flag it rather than average it away. *The three-way contrast does not carry across datasets*, as discussed under RQ4. *The epoch budget binds*: the selected checkpoint sits at epoch 92–100 everywhere and at 98–100 in nearly all $R = 256$ cells, so the models are near-converged but not converged, and the condition with the largest effects is the one most likely to be truncated. *DReG's cost is not negligible* at the scale where it helps most.

Four design limitations bound the claims. First, R is deliberately confounded with the likelihood family and with head capacity: the $R = 16$ head has $6.8\times$ the parameters of the $R = 2$ head (Table I), so RQ1 estimates the effect of the *observation-model condition*, not a likelihood-invariant causal effect of resolution. Second, the metric is a $K_{\text{eval}} = 1000$ lower bound, not an exact NLL: comparisons are fair because the estimator is common, but absolute values are optimistic by an unknown gap. Third, three seeds per cell support paired descriptive intervals and nothing stronger. Fourth, and most consequentially for RQ4, the frozen plan specified EMNIST-*Balanced* and the executed grid used EMNIST-*Letters*. The two are format-compatible and Letters is a legitimately harder, class-balanced target, so the comparison is scientifically meaningful; but it is not the preregistered one, the study's own automated audit records the run as protocol-deviating on exactly this ground, and we therefore scope RQ4 as "consistent across digits and a harder set of handwritten characters".

D. Threats to validity

Internally, the strongest protections are that all three R conditions derive from identical source bytes, that IWAE and DReG share forward samples and seed-paired initializations, and that the test set stayed sealed until after checkpoint and configuration selection, both of which used validation only; the residual internal threat is the capacity confound above. One measured offset deserves stating: test bpd runs systematically below validation bpd on MNIST (-0.0011, -0.0044 and -0.0038 bpd at $R = 2, 16, 256$), so our test split is slightly the easier of the two, and at $R \in \{16, 256\}$ that offset exceeds the DReG effect we report. It contaminates no comparison here — every contrast is paired within a single split — but absolute bpd values should not be read against numbers from a different partition. Validation and test rank the 27 configurations almost identically (Spearman $\rho = 0.999$) and select the same best configuration, so selection is not overfitting the validation split. Externally, the results cover one architecture family (small MLP, single stochastic layer, $d_z = 20$), two handwritten-character datasets and one optimizer setting. Statistically, eighteen contrasts per dataset are reported without multiplicity correction, which is defensible only because we frame them descriptively.

VIII. Conclusion

We answered four questions with a frozen, fully-crossed $3 \times 9 \times 3$ design repeated on two datasets and scored on a common bits-per-dimension scale. Raising observation complexity from binary to 16-level to grayscale changes density-estimation complexity systematically and by a large margin ($+0.47$ and $+0.98$ bpd per step), by an amount that depends mildly on training. DReG-IWAE improves on standard IWAE in 23 of 24 matched cells, but the margin is small ($\approx 10^{-3}$ bpd) and flat rather than growing in K, it is slower to its best checkpoint in every cell, and it is the less reproducible estimator at $R = 256$.

Increasing K gives consistent but sharply diminishing gains, roughly six times larger in absolute bits at 256 levels than at 2, saturating at $K = 20$ for coarse observations and not yet saturated at $K = 50$ for fine ones. Across matched contrasts, 52 of 54 agree in sign between digits and letters; only the fine-grained interaction contrasts fail to transfer.

The probabilistic insight is that the observation model is not a preprocessing detail: it sets the dispersion of the importance weights, and therefore how much a tighter bound is worth and how hard the inference gradient is to estimate. Conclusions about K calibrated on binarized MNIST understate what K is worth at realistic intensity resolution. Future work should raise the seed count in the $R = 256$ corner where effects are largest and variance highest, extend the epoch budget until those curves flatten, add a capacity-matched $R = 16$ head to break the confound identified above, and run the preregistered EMNIST-Balanced split.

Team Contribution Statement

Author 1 led the probabilistic formulation and the derivation and verification of the three observation models and the DReG estimator, and wrote Section III. **Author 2** implemented the training pipeline, pilot gate, fingerprinting infrastructure and experiment runner, and wrote Sections IV–V. **Author 3** designed and executed the statistical analysis, produced all figures and tables, and wrote Section VI. **Author 4** guided us on the whole project as an advisor.

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